

3. A manufacturer produces cylindrical shafts with a nominal diameter of 50 mm. A quality control engineer measures the diameter of five randomly selected shafts using a micrometer, and the recorded measurements are 49.85, 49.90, 49.87, 49.92, 49.89 (mm). Calculate the accuracy and precision of the measurements. Comment on whether the manufacturing process is more accurate, precise or neither based on the calculated values.

[0.75 + 0.75 + 0.5 = 2 marks]

Sol:-

Accuracy

Given True value (T) = 50 mm

Observations (X_i)

= 49.85, 49.90, 49.87, 49.92, 49.89 mm

$$\text{Accuracy } (\sigma_T) = \sqrt{\frac{\sum_{i=1}^n (X_i - T)^2}{n-1}}$$

$$= \sqrt{\frac{(49.85 - 50)^2 + (49.90 - 50)^2 + (49.87 - 50)^2 + (49.92 - 50)^2 + (49.89 - 50)^2}{4}}$$

$$= 0.1303 \quad (0.75 \text{ marks})$$

Precision

$$\text{Mean } (\bar{X}) = 49.85 + 49.90$$

$$+ 49.87 + 49.92$$

$$+ 49.89$$

$$5$$

$$= 49.886$$

$$\sigma_X = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

$$= \sqrt{\frac{(49.85 - 49.886)^2 + (49.90 - 49.886)^2 + (49.87 - 49.886)^2 + (49.92 - 49.886)^2 + (49.89 - 49.886)^2}{4}}$$

$$= 0.0270$$

(0.75 marks)

High Precision $\sigma_X = 0.0270$

Low Accuracy $\sigma_T = 0.1303$

The manufacturing process is precise but not accurate. (0.5 marks)

4. Holes of 75 ± 0.05 mm diameter are to be checked by the help of GO and NO-GO Plug gage. Design the GO and NO-GO Plug gages, sketch it and show its GO-size and NO-GO size dimensions. Assume normal wear allowance and gage maker's tolerance.

[0.75 + 0.75 + 0.5 = 2 marks]

Solution - High limit of hole = 75.05 mm

Low limit of hole = 74.95 mm

Work tolerance = $75.05 - 74.95 = 0.1$ mm

Gage maker's tolerance = 0.01 mm

Wear allowance = 0.005 mm

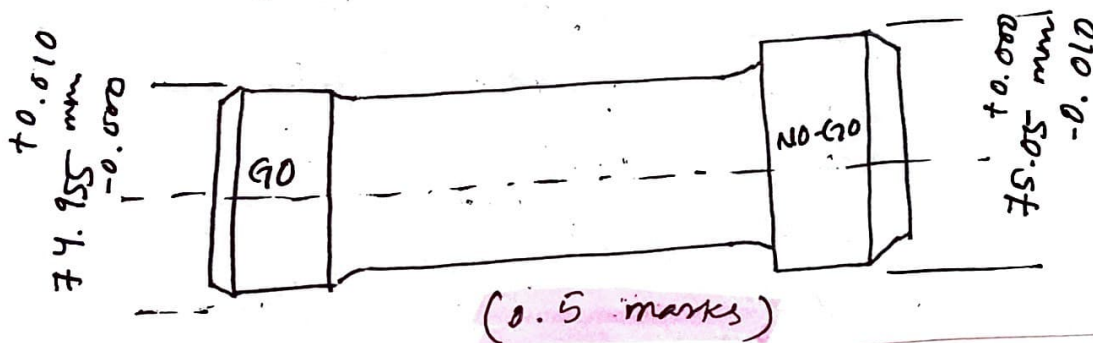
'GO' side of plug gage = L.L of hole (MMC)
= 74.95 mm

'GO' side of plug gage after the application
of wear allowance = $74.95 + 0.005$
= 74.955 mm

Dimension of plug gage are given as :

'GO' $74.955^{+0.010}_{-0.000}$ mm
(0.75 mark)

'Not GO' $75.05^{+0.000}_{-0.010}$ mm
(0.75 marks)

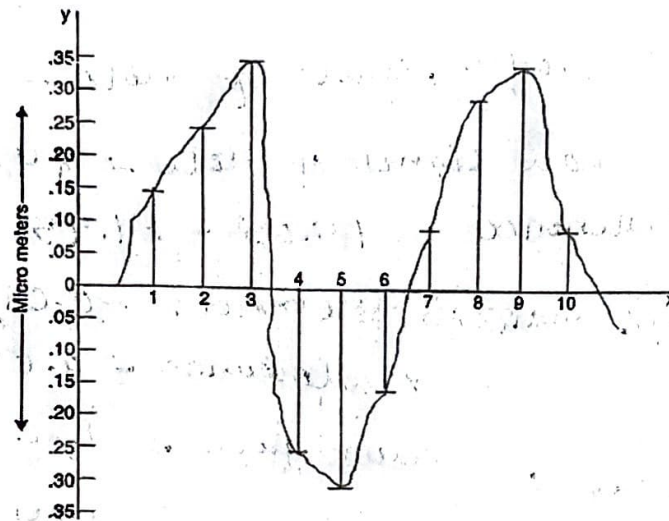


$74.955^{+0.005}_{-0.005}$
GO

$75.05^{+0.005}_{-0.005}$
NO-GO
(Bilateral)

5. Calculate the R_a and the rms values for the surface roughness shown in the figure.

[1 + 1 = 2 marks]



Solution:- The values of y and y^2 are

<u>Ordinate</u>	<u>y</u>	<u>y²</u>
1	0.15	0.0225
2	0.25	0.0625
3	0.35	0.1225
4	0.25	0.0625
5	0.30	0.0900
6	0.15	0.0225
7	0.10	0.0100
8	0.30	0.0900
9	0.35	0.1225
10	0.10	0.0100

$$\sum Y_n = 2.30, \sum Y_n^2 = 0.615$$

$$R_a = \frac{\sum Y_n}{n} = \frac{2.30}{10} = \underline{0.23 \mu m} \quad (1 \text{ marks})$$

$$rms = \sqrt{\frac{\sum Y_n^2}{n}} = \sqrt{\frac{0.615}{10}} = \underline{0.248 \mu m} \quad (1 \text{ marks})$$

6) Desired length = 150 mm
 Pattern length = 150 mm
 Contraction = 6%
 Say, machining removal = x mm

$$\therefore 150 - \frac{6 \times 150}{100} = 140 + 2x$$

$$\Rightarrow 2x = 10$$

$$\Rightarrow x = 0.5$$

$\therefore$ A thin strip of material having 0.5 mm thickness should be removed from all sides of cast product during machining.

Desired diameter = 46 mm

Pattern diameter = 50 mm

Contraction = 6%

Say, machining removal in diameter = y mm

$$\therefore 50 - \frac{6 \times 50}{100} = 46 + 2y$$

$$\Rightarrow 2y = 10$$

$$\Rightarrow y = 0.5$$

7)

Cast metal

$$V_c = 150 \times 150 \times 25 \text{ mm}^3$$

$$A_c = (2 \times 150 \times 150) + (4 \times 150 \times 25) \text{ mm}^2$$

$$\therefore t_c = K \left(\frac{V_c}{A_c} \right)^2 = K \left\{ \frac{150 \times 150 \times 25}{150 \times 25 (12 + 4)} \right\}^2$$

$$= K \left(\frac{150}{16} \right)^2 = 87.89 K$$

Riser part

$$V_R = a^3$$

$$A_R = 5a^2$$

} Say, side length of the cubic riser is a

$$\therefore t_R = K \left(\frac{V_R}{A_R} \right)^2 = K \left(\frac{a^3}{5a^2} \right)^2 = \frac{a^2}{25} K$$

$$\text{Now, } t_R = 1.3 t_c$$

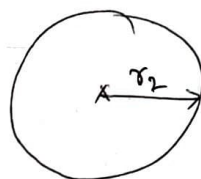
$$\Rightarrow \frac{a^2}{25} K = 1.3 \times 87.89 K \Rightarrow a = 53.44 \text{ mm}$$

8)



$$V_1 = \frac{4}{3} \pi r_1^3$$

$$A_1 = 4 \pi r_1^2$$



$$V_2 = \frac{4}{3} \pi r_2^3$$

$$A_2 = 4 \pi r_2^2$$

$$\text{Material is same } \Rightarrow V_2 = 8 V_1 \Rightarrow r_2 = 2 r_1$$

$$\therefore \frac{t_2}{t_1} = \left[\frac{\frac{\frac{4}{3} \pi r_2^3}{4 \pi r_2^2}}{\frac{\frac{4}{3} \pi r_1^3}{4 \pi r_1^2}} \right]^2 = \left[\frac{\frac{r_2}{3}}{\frac{r_1}{3}} \right]^2 = \left(\frac{r_2}{r_1} \right)^2 = \left(\frac{2 r_1}{r_1} \right)^2 = 4$$

$$\therefore 20 \times 4 = 80 \text{ min.}$$

Model Answers Part-III

Given,

$$W = 300 \text{ mm},$$

$$t_o = 25 \text{ mm}$$

$$t_f = 22 \text{ mm}$$

$$R = 250 \text{ mm}$$

$$N = 50 \text{ kN/mm}$$

$$K = 415 \text{ MPa}, \eta = 0.15, \mu = 0.12$$

$$d = t_o - t_f = 25 - 22 = 3 \text{ mm}$$

$$d_{\max} = \mu^2 R = (0.12)^2 \times 250 = 3.6 \text{ mm} \checkmark$$

As $d_{\max} > d$ The friction is sufficient to permit the rolling operation.

$$\text{Roll force} = y_f \times W \times L$$

$$\left[\epsilon = \ln \frac{t_o}{t_f} \right], L = \sqrt{R \times d}$$

$$y_f = \frac{K \epsilon^\eta}{1 + \eta} = \frac{415 \times 10^6 \left[\ln \frac{25}{22} \right]^{0.15}}{1 + 0.15} = 265.06 \times 10^6 \text{ Pa}$$

$$\text{Roll force} = 265.06 \times 10^6 \times 300 \times \sqrt{250 \times 3} \times 10^{-6}$$

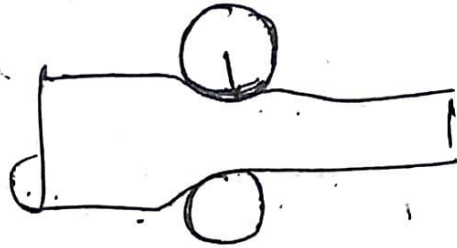
$$= 2177690.116 \text{ N}$$

$$\text{Torque} = 0.5 \times F \times L = 0.5 \times 2177690.116 \times \sqrt{250 \times 3} \times 10^{-3}$$

$$= 29819.24 \text{ Nm}$$

Part III

$$\tau = \vec{r} \times \vec{F}$$



~~Power~~
$$Power = 2\pi N \times F \times L$$

$$= 2\pi \times \frac{50}{60} \times 2177690.116 \times \sqrt{250 \times 3} \times 10^{-3}$$

$$= 312108.1499 \text{ W}$$

$$Horsepower = \frac{312108.1499}{745.7} = 418.54 \text{ Horsepower}$$

$$\therefore \text{Roll force} = 2177690.116 \text{ N, Torque} = 29819.24 \text{ Nm}$$

$$Horsepower = 418.54 \text{ Horsepower}$$

11.) Given,

$$h_0 = 75 \text{ mm, } D_0 = 50 \text{ mm}$$

$$h_f = 36 \text{ mm, } h_1 = 62 \text{ mm, } h_2 = 49 \text{ mm}$$

$$K = 415 \text{ MPa, } \eta = 0.17, \mu = 0.1$$

Force at height $h_f = 36 \text{ mm}$

$$\epsilon = \ln \frac{h_0}{h_f} = \ln \frac{75}{36} = 0.733$$

$$A = \frac{\pi D_0^2}{4} = 1962.5 \text{ mm}^2$$

$$y_f = K \ln K e^\eta = 415 \times 10^6 \times (0.733)^{0.17} = 393.65 \text{ MPa}$$

$$K_f = 1 + \frac{0.4 \times 0.1 \times 60}{75} = 1.02$$

$$F = K_f \times y_f \times A = 1.02 \times 393.65 \times 10^6 \times 1962.5 \times 10^{-6}$$

$$= 787988.8875 \text{ N}$$

Force at height $h_1 = 62 \text{ mm}$

assuming volume remains constant,

$$v_1 = v_2$$

$$\Rightarrow \frac{4}{3} \pi \frac{D_1^3 h_1}{4} = \frac{4}{3} \pi \frac{D_2^3 h_2}{4} \Rightarrow D_0^3 h_0 = D_2^3 h_2$$

$$\Rightarrow D_2 = \sqrt[3]{\frac{h_0}{h_2}} \times D_0$$

$$= \sqrt[3]{\frac{75}{62}} \times 50$$

$$D_2 = 54.99 \text{ mm}$$

$$\epsilon = \ln \frac{h_0}{h_1} = \ln \frac{75}{62} = 0.19$$

$$y_f = 415 \times 10^6 \times (0.19)^{0.17} = 312.92 \text{ MPa}$$

$$A = \frac{\pi D_1^2}{4} = 2373.76 \text{ mm}^2$$

$$K_f = 1 + \frac{0.4 \times 0.1 \times 54.99}{62} = 1.035$$

$$F = K_f \times y_f \times A = 312.92 \times 10^6 \times 1.035 \times 2373.76 \times 10^{-6} \\ = 768794.8735 \text{ N}$$

For height at 49 mm,

$$V_1 = V_2$$

$$D_2 = \sqrt{\frac{h_1}{h_2}} \times D_1 = \sqrt{\frac{75}{49}} \times 50 = 61.85 \text{ mm}$$

$$A = \frac{\pi D_2^2}{4} = \frac{3.14 \times 61.85^2}{4} = 3002.95 \text{ mm}^2$$

$$K_f = 1 + 0.4 \times 0.1 \times \frac{61.85}{49} = 1.05$$

$$y_f = K E^n = 415 \times \left(\ln \frac{75}{49} \right)^{0.17} = 358.91 \text{ MPa}$$

$$F = y_f \times K_f \times A$$

$$= 358.91 \times 10^6 \times 1.05 \times 3002.95 \times 10^{-6} \\ = 1131678 \text{ N}$$

$$\epsilon = \ln r_2 = \ln 4 = 1.386$$

$$\text{extrusion strain} = a + b \epsilon = 0.8 + 1.5 \ln 4 = 2.879$$

$$\bar{y}_f = \frac{K (\ln r_2)^n}{1+n} = \frac{350 \times (1.386)^{0.18}}{1.18} = 314.55 \text{ MPa}$$

$$P = \bar{y}_f \left(\epsilon + \frac{2L}{D_1} \right) = 314.55 \times 10^6 \left[2.879 + 2 \times \frac{75}{25} \right]$$

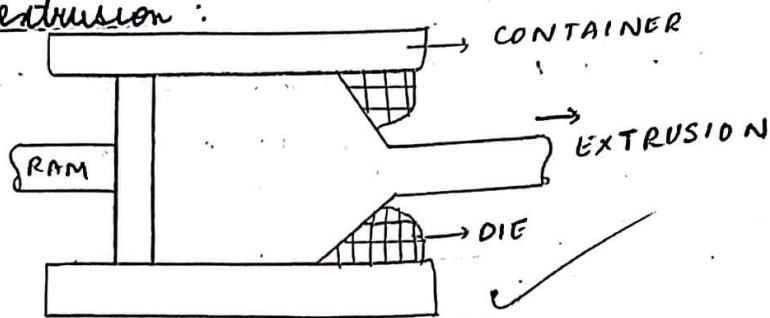
$$P = 2792.88 \text{ MPa}$$

also find (P) at the end of blister

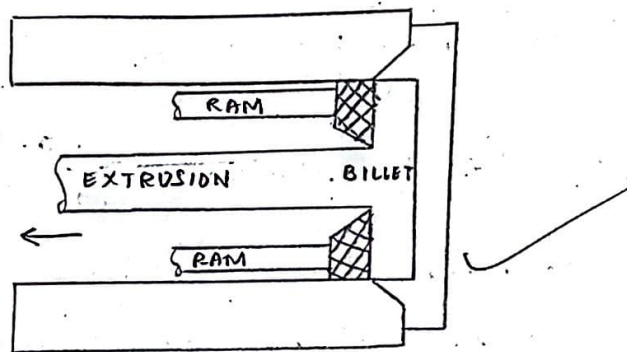
$$\text{for } L=0, P = 314.572 \left(2.879 + 2 \times \frac{0}{25} \right) \\ = 905.1810 \text{ MPa}$$

- 3) Different extrusion processes are
- i) Direct extrusion
 - ii) Indirect extrusion
 - iii) Hollow extrusion
 - iv) Hydrostatic extrusion

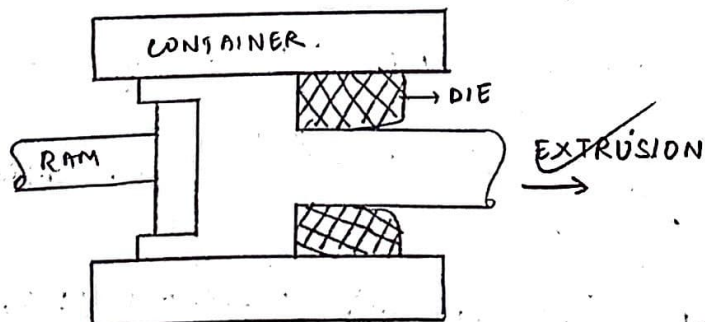
Direct extrusion:



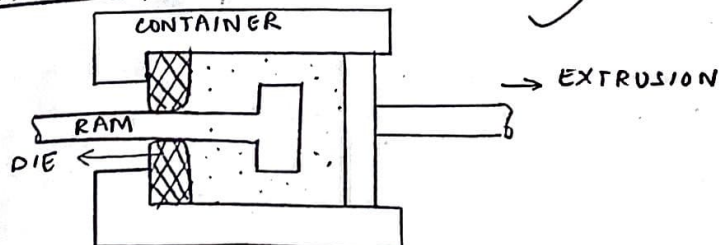
Indirect Extrusion:



Hollow extrusion



Hydrostatic extrusion



24) Difference between Hot and cold working conditions:

Hot Working condition	Cold Working condition
• The hot working condition is operated at a temperature greater than the recrystallisation temperature of the metal ✓ • Provides poor dimensional accuracy ✓ • Provides poor surface finish ✓ • No strain hardening effect can be observed ✓ • Larger deformation in the metal workpiece can be observed ✓	• The cold working condition is operated at a temperature lesser than the recrystallisation temperature of the metal. ✓ • Provides better dimensional accuracy ✓ • Provides better surface finish ✓ • Strain hardening effect can be observed ✓ • Relatively smaller deformation is achieved in the metal workpiece. ✓

(Additional Point)